Introduction to Limits

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Evaluating limits**

Evaluate $\lim_{x\to 2} \frac{x^2-4}{x-2}$ by first simplifying the expression.

$\frac{(x-2)(x+2)}{x-2} = x+2$ for $x \neq 2$, so the limit is 4.

2 **[Foundational]** • **Target: Evaluating limits**

Evaluate by direct substitution where possible:

a $\lim_{x\to 3} (x^2+2x-1) = 14$ **b** $\lim_{x\to 1} \frac{x+5}{x+1} = 3$

(a) $= 14$; (b) $= 3$

3 **[Foundational]** • **Target: Evaluating limits**

Using a table of values near $x = 0$ (in radians), estimate $\lim_{x\to 0} \frac{\sin x}{x}$.

Values approach 1 from both sides; the limit is 1.

4 **[Foundational]** • **Target: One-sided limits and limits at infinity**

Let $f(x) = \frac{|x|}{x}$. Find:

a $\lim_{x\to 0^+} f(x) = 1$ **b** $\lim_{x\to 0^-} f(x) = -1$ **c** $\lim_{x\to 0} f(x)$ does not exist

(a) $= 1$; (b) $= -1$; (c) does not exist

5 **[Foundational]** • **Target: One-sided limits and limits at infinity**

Evaluate each limit at infinity:

a $\lim_{x\to\infty} \frac{3x^2+5}{x^2-2} = 3$ **b** $\lim_{x\to\infty} \frac{2x+1}{x^2+4} = 0$

(a) $= 3$; (b) $= 0$

6 **[Foundational]** • **Target: One-sided limits and limits at infinity**

The graph of $f(x) = \frac{1}{x-2} + 3$ has asymptotes $x = 2$ and $y = 3$. Express each asymptote as a limit statement.

$\lim_{x\to 2^+} f(x) = +\infty$ (and $\lim_{x\to 2^-} f(x) = -\infty$); $\lim_{x\to\pm\infty} f(x) = 3$.

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7 [Foundational] • Target: More limit computations

Evaluate $\lim_{x\to 4}\frac{\sqrt{x}-2}{x-4}$ by rationalizing the numerator.

Multiply by $\frac{\sqrt{x}+2}{\sqrt{x}+2}$: $\frac{x-4}{(x-4)(\sqrt{x}+2)}=\frac{1}{\sqrt{x}+2}\to\frac{1}{4}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: More limit computations

Evaluate $\lim_{x\to -1}\frac{x^3+1}{x+1}$ by factoring (using the sum-of-cubes formula).

$x^3+1=(x+1)(x^2-x+1)$, so the limit equals $\lim_{x\to -1}(x^2-x+1)=1+1+1=3$.

9 [Intermediate] • Target: More limit computations

Evaluate $\lim_{x\to 0}\frac{\tan x}{x}$, using $\lim_{x\to 0}\frac{\sin x}{x}=1$ and $\tan x=\frac{\sin x}{\cos x}$.

$\frac{\tan x}{x}=\frac{\sin x}{x}\cdot\frac{1}{\cos x}\to 1\cdot\frac{1}{1}=1$.

10 [Intermediate] • Target: Continuity and limits together

A function f is defined by $f(x)=x+3$ for $x\neq 2$, and $f(2)=10$. Find $\lim_{x\to 2}f(x)$ and state whether f is continuous at $x=2$.

$\lim_{x\to 2}f(x)=2+3=5$. Since $f(2)=10\neq 5$, f is not continuous at $x=2$ — it has a removable discontinuity.

11 [Intermediate] • Target: Continuity and limits together

Estimate $\lim_{x\to\infty}\left(\sqrt{x^2+1}-x\right)$ by rationalizing.

Multiply by $\frac{\sqrt{x^2+1}+x}{\sqrt{x^2+1}+x}$: $\frac{1}{\sqrt{x^2+1}+x}\to 0$ as $x\to\infty$.

12 [Intermediate] • Target: Limits from graphs and tables

A function g satisfies $g(1.9)=5.8$, $g(1.99)=5.98$, $g(1.999)=5.998$, $g(2.1)=6.2$, $g(2.01)=6.02$, $g(2.001)=6.002$. Based on this table, estimate $\lim_{x\to 2}g(x)$.

Both sides approach 6: $\lim_{x\to 2}g(x)=6$.

13 [Intermediate] • Target: Limits from graphs and tables

Explain, using one-sided limits, why $\lim_{x\to 0}\frac{1}{x^2}$ exists as an infinite limit (both one-sided limits agree), while $\lim_{x\to 0}\frac{1}{x}$ does not exist even as an infinite limit.

For $\frac{1}{x^2}$: as $x\to 0^+$ or $x\to 0^-$, $x^2>0$ always, so both one-sided limits are $+\infty$ — they agree, so we say $\lim_{x\to 0}\frac{1}{x^2}=+\infty$. For $\frac{1}{x}$: as $x\to 0^+$, $\frac{1}{x}\to+\infty$, but as $x\to 0^-$, $\frac{1}{x}\to-\infty$ — the one-sided limits disagree (not even both infinite in the same direction), so the limit does not exist in any sense.

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14 [Intermediate] • Target: Visualizing a limit graphically

The graph of $f(x) = \frac{x^2-4}{x-2}$ (which simplifies to $x+2$ with a hole at $x=2$) is shown. Use the graph to estimate $\lim_{x\to 2} f(x)$, and explain why the hole doesn't affect the limit's value.

The graph approaches the height 4 from both sides as $x \to 2$, even though the function is undefined exactly at $x = 2$ (the hole). A limit only depends on the function's behavior NEAR the point, not its value (or lack of one) AT the point, so $\lim_{x\to 2} f(x) = 4$ regardless of the hole.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: The formal idea of a limit (intuitive) • Cross-Lesson: 4.01 Circles and Parabolas

Explain, in your own words, what it means for $\lim_{x\to a} f(x) = L$, without using formal epsilon-delta language.

It means that as x gets arbitrarily close to a (from either side, without necessarily equaling a), the function's output values $f(x)$ get arbitrarily close to the single number L.

16 [Challenging] • Target: Limit laws • Cross-Lesson: 4.02 Ellipses

Given $\lim_{x\to 3} f(x) = 5$ and $\lim_{x\to 3} g(x) = -2$, use the limit laws to evaluate:

a $\lim_{x\to 3}[f(x)+g(x)]$ $= 5 + (-2) = 3$

b $\lim_{x\to 3}[f(x)\cdot g(x)]$ $= 5(-2) = -10$

c $\lim_{x\to 3} \frac{f(x)}{g(x)}$ $= \frac{5}{-2} = -\frac{5}{2}$

(a) $= 5 + (-2) = 3$; (b) $= 5(-2) = -10$; (c) $= \frac{5}{-2} = -\frac{5}{2}$

17 [Challenging] • Target: Limits involving piecewise functions • Cross-Lesson: 4.03 Hyperbolas

A function is defined by $f(x) = x^2$ for $x < 1$ and $f(x) = 3x - 1$ for $x \geq 1$. Find $\lim_{x\to 1^-} f(x)$, $\lim_{x\to 1^+} f(x)$, and state whether $\lim_{x\to 1} f(x)$ exists.

$\lim_{x\to 1^-} f(x) = 1^2 = 1$. $\lim_{x\to 1^+} f(x) = 3(1) - 1 = 2$. Since the one-sided limits disagree ($1 \neq 2$), $\lim_{x\to 1} f(x)$ does NOT exist.

18 [Challenging] • Target: Limits from graphs and tables • Cross-Lesson: 5.01 Vectors: Components and Operations

Consider the rational function $f(x) = \frac{ax^2+bx+c}{dx^2+ex+f}$, where a, b, c, d, e, f are non-zero constants. Given that $\lim_{x\to\infty} f(x) = \frac{1}{2}$ and $\lim_{x\to -1} f(x) = \infty$, determine the value of $\frac{a}{d}$.

Final Answer: $\frac{1}{2}$. For a rational function where the degree of the numerator is equal to the degree of the denominator, the limit as $x \to \infty$ is the ratio of the leading coefficients. Therefore, $\lim_{x\to\infty} f(x) = \frac{a}{d}$. We are given that this limit is $\frac{1}{2}$, so $\frac{a}{d} = \frac{1}{2}$. The condition $\lim_{x\to -1} f(x) = \infty$ implies that $x = -1$ is a vertical asymptote. This means that the denominator is zero when $x = -1$, and the numerator is.

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19 [Challenging] • Target: Visualizing a limit graphically • Cross-Lesson: 5.02 The Dot Product

Consider the rational function $f(x) = \frac{x^2 - 9}{x - 3}$. What is the value of $f(x)$ as x approaches 3? Show your complete algebraic work.

Final Answer: 6. The function is given by $f(x) = \frac{x^2 - 9}{x - 3}$. We can factor the numerator as a difference of squares: $x^2 - 9 = (x - 3)(x + 3)$. Thus, for $x \neq 3$, we can simplify the function to $f(x) = \frac{(x - 3)(x + 3)}{x - 3} = x + 3$. To find the limit as x approaches 3, we can substitute 3 into the simplified expression: $\lim_{x \to 3} f(x) = \lim_{x \to 3} (x + 3) = 3 + 3 = 6$. Although the function is undefined at $x = 3$, the limit exists and is 6. Therefore, as x approaches 3.

20 [Challenging] • Target: The formal idea of a limit (intuitive) • Cross-Lesson: 5.03 Parametric Equations

Consider the rational function $f(x) = \frac{x^2 - 4}{x - 2}$. What is the value of the function as x approaches 2? Show your complete algebraic work.

Final Answer: 4. The given function is $f(x) = \frac{x^2 - 4}{x - 2}$. We can factor the numerator as a difference of squares: $x^2 - 4 = (x - 2)(x + 2)$. Therefore, $f(x) = \frac{(x - 2)(x + 2)}{x - 2}$. For $x \neq 2$, we can cancel the $(x - 2)$ term, so $f(x) = x + 2$. As x approaches 2, the value of $f(x)$ approaches $2 + 2 = 4$. Although the function is technically undefined at $x = 2$ due to division by zero, the limit as x approaches 2 exists.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Intuitive Concept and Numerical Estimates

What does $\lim_{x \to c} f(x) = L$ mean intuitively?

a. **As x gets arbitrarily close to c (from either side, $x \neq c$), $f(x)$ gets arbitrarily close to L** ✓

b. $f(c) = L$ always

c. The slope of f at c is L

d. $f(x)$ is continuous at c

A limit describes the value that $f(x)$ approaches as x nears c, completely independent of what happens at $x = c$ itself.

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22 [Foundational] • Target: Intuitive Concept and Numerical Estimates

Use numerical estimation with $x = 1.9, 1.99, 2.01, 2.1$ to estimate $\lim_{x \to 2} (3x - 1)$.

a. 5 ✓

b. 6

c. 4

d. 3

$3(1.99) - 1 = 4.97$; $3(2.01) - 1 = 5.03$. Values converge toward 5 as $x \to 2$.

23 [Foundational] • Target: Intuitive Concept and Numerical Estimates

If $f(3)$ is undefined, can $\lim_{x \to 3} f(x)$ still exist?

a. **Yes, because limits describe behavior near $x = 3$, not at $x = 3$** ✓

b. No, limits require $f(c)$ to be defined

c. Only if $f(x) = 0$

d. Only for polynomials

The existence of a limit at c depends entirely on values in a punctured neighborhood around c, regardless of whether $f(c)$ is defined.

24 [Foundational] • Target: Graphical Limits and One-Sided Limits

When does the two-sided limit $\lim_{x \to c} f(x)$ exist?

a. **If and only if both one-sided limits exist and are equal ($\lim_{x \to c^-} f(x) = \lim_{x \to c^+} f(x) = L$)** ✓

b. Whenever $f(c)$ exists

c. Whenever $\lim_{x \to c^+} f(x)$ exists

d. If f is defined on $[c, \infty)$

The two-sided limit exists if and only if both the left-hand limit and the right-hand limit exist and agree on a common value L.

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25 [Foundational] • Target: Graphical Limits and One-Sided Limits

For the step function $f(x)$ defined as 2 for $x < 1$ and 5 for $x \geq 1$, find $\lim_{x \to 1^-} f(x)$ and $\lim_{x \to 1^+} f(x)$.

a. **Left limit $= 2$, Right limit $= 5$** ✓

b. Left limit $= 5$, Right limit $= 2$

c. Both equal 2

d. Both equal 5

As $x \to 1^-$ (from the left), $f(x) = 2$. As $x \to 1^+$ (from the right), $f(x) = 5$.

26 [Foundational] • Target: Graphical Limits and One-Sided Limits

Does the two-sided limit $\lim_{x \to 1} f(x)$ exist for the step function in the previous question?

a. **No, because the left limit (2) does not equal the right limit (5)** ✓

b. Yes, it equals 5

c. Yes, it equals 3.5

d. Yes, it equals 2

Because $\lim_{x \to 1^-} f(x) = 2 \neq 5 = \lim_{x \to 1^+} f(x)$, the two-sided limit does not exist (DNE).

27 [Foundational] • Target: Algebraic Evaluation of Limits

Evaluate $\lim_{x \to 4} \frac{x^2 - 16}{x - 4}$.

a. 8 ✓

b. 0

c. 4

d. Does not exist

Indeterminate form $0/0$. Factor: $\frac{(x-4)(x+4)}{x-4} = x + 4$. As $x \to 4$, $4 + 4 = 8$.

28 [Foundational] • Target: Algebraic Evaluation of Limits

Evaluate $\lim_{x \to 0} \frac{\sqrt{x+9} - 3}{x}$.

a. $1/6$ ✓

b. $1/3$

c. 0

d. $1/9$

Multiply numerator and denominator by conjugate $(\sqrt{x+9} + 3)$: $\frac{(x+9)-9}{x(\sqrt{x+9}+3)} = \frac{1}{\sqrt{x+9}+3} \to \frac{1}{3+3} = \frac{1}{6}$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Algebraic Evaluation of Limits**

Evaluate $\lim_{x\to -2} \frac{x^3+8}{x+2}$.

a. 12 ✓

b. 0

c. 4

d. 8

Sum of cubes: $\frac{(x+2)(x^2-2x+4)}{x+2} = x^2-2x+4$. At $x=-2$: $(-2)^2-2(-2)+4 = 4+4+4 = 12$.

30 **[Intermediate]** • **Target: Limits Involving Infinity**

Evaluate $\lim_{x\to\infty} \frac{5x^2-2x}{2x^2+7x-1}$.

a. 5/2 ✓

b. ∞

c. 0

d. -2

Divide by x^2: $\lim_{x\to\infty} \frac{5-2/x}{2+7/x-1/x^2} = \frac{5}{2}$.

31 **[Intermediate]** • **Target: Limits Involving Infinity**

Evaluate $\lim_{x\to\infty} \frac{3x+1}{x^2+5}$.

a. 0 ✓

b. 3

c. ∞

d. 1/5

The degree of the denominator is higher than that of the numerator, so the limit as $x\to\infty$ is 0.

32 **[Intermediate]** • **Target: Limits Involving Infinity**

Evaluate $\lim_{x\to 3^+} \frac{1}{x-3}$ and $\lim_{x\to 3^-} \frac{1}{x-3}$.

a. $+\infty$ **and** $-\infty$ ✓

b. $+\infty$ and $+\infty$

c. $-\infty$ and $+\infty$

d. 0 and 0

As $x\to 3^+$, $x-3>0$, so $1/(x-3)\to +\infty$. As $x\to 3^-$, $x-3<0$, so $1/(x-3)\to -\infty$.

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33 [Intermediate] • Target: Limits Involving Infinity

Using the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$, find $f'(x)$ for $f(x) = 4x^2 - x + 4$.

a. $4x^2 - 1$

b. $8x + 4$

c. $4x - 1$

d. $8x - 1$ ✓

Expand $f(x+h) - f(x) = 4(2xh + h^2) - 1h = (8x - 1)h + 4h^2$. Dividing by h: $\frac{f(x+h) - f(x)}{h} = 8x - 1 + 4h$. Letting $h \to 0$: $f'(x) = 8x - 1$.

34 [Intermediate] • Target: Evaluating limits

Using the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$, find $f'(x)$ for $f(x) = 3x^2 - 3x + 3$.

a. $6x - 3$ ✓

b. $6x + 3$

c. $3x^2 - 3$

d. $3x - 3$

Expand $f(x+h) - f(x) = 3(2xh + h^2) - 3h = (6x - 3)h + 3h^2$. Dividing by h: $\frac{f(x+h) - f(x)}{h} = 6x - 3 + 3h$. Letting $h \to 0$: $f'(x) = 6x - 3$.

35 [Intermediate] • Target: One-sided limits and limits at infinity

Using the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$, find $f'(x)$ for $f(x) = 3x^2 + 4x + 5$.

a. $3x + 4$

b. $3x^2 + 4$

c. $6x + 5$

d. $6x + 4$ ✓

Expand $f(x+h) - f(x) = 3(2xh + h^2) + 4h = (6x + 4)h + 3h^2$. Dividing by h: $\frac{f(x+h) - f(x)}{h} = 6x + 4 + 3h$. Letting $h \to 0$: $f'(x) = 6x + 4$.

36 [Intermediate] • Target: More limit computations

Using the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$, find $f'(x)$ for $f(x) = 3x^2 - 2x + 5$.

a. $3x - 2$

b. $6x - 2$ ✓

c. $3x^2 - 2$

d. $6x + 5$

Expand $f(x+h) - f(x) = 3(2xh + h^2) - 2h = (6x - 2)h + 3h^2$. Dividing by h: $\frac{f(x+h) - f(x)}{h} = 6x - 2 + 3h$. Letting $h \to 0$: $f'(x) = 6x - 2$.

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37 [Intermediate] • Target: Continuity and limits together

Using the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$, find $f'(x)$ for $f(x) = 2x^2 + 2x - 1$.

a. $4x + 2$ ✓

b. $4x - 1$

c. $2x^2 + 2$

d. $2x + 2$

Expand $f(x+h) - f(x) = 2(2xh + h^2) + 2h = (4x+2)h + 2h^2$. Dividing by h: $\frac{f(x+h) - f(x)}{h} = 4x + 2 + 2h$. Letting $h \to 0$: $f'(x) = 4x + 2$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Limits from graphs and tables • Cross-Lesson: 5.01 Vectors: Components and Operations

Using the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$, find $f'(x)$ for $f(x) = 4x^2 + 2x - 4$.

a. $4x^2 + 2$

b. $8x - 4$

c. $8x + 2$ ✓

d. $4x + 2$

Expand $f(x+h) - f(x) = 4(2xh + h^2) + 2h = (8x+2)h + 4h^2$. Dividing by h: $\frac{f(x+h) - f(x)}{h} = 8x + 2 + 4h$. Letting $h \to 0$: $f'(x) = 8x + 2$.

39 [Challenging] • Target: Visualizing a limit graphically • Cross-Lesson: 5.02 The Dot Product

Using the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$, find $f'(x)$ for $f(x) = 1x^2 + 4x - 5$.

a. $1x + 4$

b. $1x^2 + 4$

c. $2x + 4$ ✓

d. $2x - 5$

Expand $f(x+h) - f(x) = 1(2xh + h^2) + 4h = (2x+4)h + 1h^2$. Dividing by h: $\frac{f(x+h) - f(x)}{h} = 2x + 4 + 1h$. Letting $h \to 0$: $f'(x) = 2x + 4$.

40 [Challenging] • Target: The formal idea of a limit (intuitive) • Cross-Lesson: 5.03 Parametric Equations

Using the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$, find $f'(x)$ for $f(x) = 2x^2 - x + 0$.

a. $2x - 1$

b. $2x^2 - 1$

c. $4x + 0$

d. $4x - 1$ ✓

Expand $f(x+h) - f(x) = 2(2xh + h^2) - 1h = (4x-1)h + 2h^2$. Dividing by h: $\frac{f(x+h) - f(x)}{h} = 4x - 1 + 2h$. Letting $h \to 0$: $f'(x) = 4x - 1$.

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41 [Challenging] • Target: Limit laws • Cross-Lesson: 6.01 Polar Coordinates and Conversions

Using the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$, find $f'(x)$ for $f(x) = 1x^2 - 3x - 1$.

a. $2x - 3$ ✓

b. $1x - 3$

c. $2x - 1$

d. $1x^2 - 3$

Expand $f(x+h) - f(x) = 1(2xh + h^2) - 3h = (2x - 3)h + 1h^2$. Dividing by h: $\frac{f(x+h) - f(x)}{h} = 2x - 3 + 1h$. Letting $h \to 0$: $f'(x) = 2x - 3$.

42 [Challenging] • Target: Limits involving piecewise functions • Cross-Lesson: 6.02 Polar Equations and Graphs

Using the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$, find $f'(x)$ for $f(x) = 4x^2 + x - 3$.

a. $4x^2 + 1$

b. $8x - 3$

c. $4x + 1$

d. $8x + 1$ ✓

Expand $f(x+h) - f(x) = 4(2xh + h^2) + 1h = (8x + 1)h + 4h^2$. Dividing by h: $\frac{f(x+h) - f(x)}{h} = 8x + 1 + 4h$. Letting $h \to 0$: $f'(x) = 8x + 1$.

43 [Challenging] • Target: Intuitive Concept and Numerical Estimates • Cross-Lesson: 7.01 Exponential Functions and Their Graphs

Using the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$, find $f'(x)$ for $f(x) = 3x^2 + x - 5$.

a. $6x - 5$

b. $6x + 1$ ✓

c. $3x + 1$

d. $3x^2 + 1$

Expand $f(x+h) - f(x) = 3(2xh + h^2) + 1h = (6x + 1)h + 3h^2$. Dividing by h: $\frac{f(x+h) - f(x)}{h} = 6x + 1 + 3h$. Letting $h \to 0$: $f'(x) = 6x + 1$.

44 [Challenging] • Target: Graphical Limits and One-Sided Limits • Cross-Lesson: 7.02 Logarithms and Their Properties

Using the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$, find $f'(x)$ for $f(x) = 3x^2 - 4x + 3$.

a. $3x - 4$

b. $3x^2 - 4$

c. $6x + 3$

d. $6x - 4$ ✓

Expand $f(x+h) - f(x) = 3(2xh + h^2) - 4h = (6x - 4)h + 3h^2$. Dividing by h: $\frac{f(x+h) - f(x)}{h} = 6x - 4 + 3h$. Letting $h \to 0$: $f'(x) = 6x - 4$.

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45 [Challenging] • Target: Algebraic Evaluation of Limits • Cross-Lesson: 7.03 Exponential and Logarithmic Equations

Using the limit definition $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$, find $f'(x)$ for $f(x) = 1x^2 + 3x - 5$.

a. $1x^2 + 3$

b. $2x - 5$

c. $1x + 3$

d. $2x + 3$ ✓

Expand $f(x+h) - f(x) = 1(2xh + h^2) + 3h = (2x+3)h + 1h^2$. Dividing by h: $\frac{f(x+h) - f(x)}{h} = 2x + 3 + 1h$. Letting $h \to 0$: $f'(x) = 2x + 3$.

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